AI × CYBERSECURITY · SEMESTER PROJECT · MUST UNIVERSITY

Voice-Spoofing Detection for Fraud Prevention

Execution Plan v3.0: from the midterm to delivery, with model v2, live analysis and production, stage by stage, and who owns what

TEAM
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REPOSITORY
github.com/kahlanzzi/voice-spoofing-detection

VERSION / DATE
v3.0 · 10 November 2026 (replaces v2.1 and the unadopted v2.2 draft)

CONTROLLING DOCUMENTS
Decision register (D1–D55) › SRS v3.0 (v4.0 in preparation) › SDD v1.0 (v2.0 in preparation) › Protocol v1.0 + addendum › this plan › Proposal

Where we are today

DONE **Stages 0–10**	Repository, audit, threat model, Protocol v1.0 (F1), split and freeze (F2), preprocessing (F3), baseline, CNN (3 seeds), risk policy on val-B. Briefs 00–10 accepted.
DONE **Mock MVP**	Stage 17 API and Stage 18 React app merged (PRs #77–#109): sign-in, upload, result, History, status, Docker/nginx. Runs on a labelled MockAnalyzer.
DONE **Stage 11 ready**	D46 run procedure approved (PRs #67, #112); runner merged (PR #68).
NEW **After the midterm**	Professor's feedback (9 Nov) turned into D48–D55, approved in PR #111: model v2, one joint external run, live analysis, production infrastructure, UI redesign.
NEXT **This week**	Stage 11 PR B and PR C · D50 corpora and augmentation · UI directions · PostgreSQL and process pool · Stage 13 runner for three model sets.
DATE **Hard dates**	v2 freeze (F5) cut-off: Fri 4 Dec 2026, 23:59 Tunis (D48). Delivery target: Wed 23 Dec 2026, to be confirmed against the course calendar.

Every fact in this plan comes from the merged repository (decision register, project log, protocol) at `eb16a85` , PR #68 merged.

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1. How to use this plan

This is the team's working reference from the midterm to delivery. It says what we build, in which order, by when, and what "done" means at each step. It does not repeat the rules in full: the decision register, the protocol and `AGENTS.md` hold them, and this plan points to them.

- **Start of a step:** read its card (Sections 8–12): goal, tasks, outputs, gate, dates and the decisions that apply.
- **During a step:** follow the hard rules (Section 4) and the workflow (Section 5).
- **End of a step:** done only when its gate is passed: outputs merged into `main` through a reviewed PR with green CI, and the project log updated.
- **Late or blocked?** Say it in the weekly sync the same day. The critical path (Section 7) shows which delays move the 4 December cut-off and which do not.
- **Something conflicts?** The decision register wins, then SRS, then SDD, then the protocol, then this plan. Never resolve a conflict silently.

> **Why v3.0?.** v2.1 planned one CNN, an upload-only service and a DEEP-VOICE run right after the v1 freeze. The midterm review (9 November) asked for a stronger model, augmentation, live analysis, production infrastructure and a better interface. D48–D55 decide how, without breaking the rules that protect the results. v3.0 turns those decisions into owners and dates. The v2.2 draft was never adopted; its status content is folded in here.

2. What changed after the midterm

The professor approved the methodology (branches, commits, PRs, decisions) and raised the points below. Each one is now an approved decision (PR #111, 10 November 2026).

Professor's point	Decision	What we do
The platform cannot be tested: it runs on mock data	D48	Finish v1 to its freeze (F4) and replace the mock with the real model as soon as F4 is verified.
Move beyond the CNN: LSTM or voice-specific models	D49	Three pre-declared v2 candidates: `cnn-aug` (control), `crnn` (CNN + BiLSTM), `aasist` (spoofing-specific).
FoR is too clean: augment for real-world audio	D50	On-the-fly augmentation (noise, rooms, codecs, gain, RawBoost) for REAL and FAKE alike, disjoint from Stage 14.
Show and justify the confidence-score decisions	D55	A decision board in the UI: calibrated p, threshold, uncertain band and how each was derived.
Live check during calls, with free tools	D51, D52	Live microphone mode and `/api/live` (Must); Chrome extension for web calls (Should).
Infrastructure: CPU/GPU, database, bottlenecks, justified limits	D53	CPU process pool, `503` + `Retry-After` when busy, PostgreSQL, limits from a load test.

Professor's point	Decision	What we do
Staging before production	D54	dev → staging (auto after green CI) → production (release tag + reviewer).
The UI looks generic	D55	Three design directions, one chosen, moderated usability test (SUS ≥ 68).

The most important consequence: **Stage 13 now waits for v2** (D48). DEEP-VOICE can be used once, and D35 makes that run dataset-wide, so the only way to compare v1 and v2 on it is one joint run. If v2 is not frozen by **Friday 4 December, 23:59 Tunis**, Stage 13 runs with v1 and the baseline only, and v2 never gets an external result.

3. The project on one page

What we build

A **decision-support** security control that analyses speech, from an uploaded recording or live during a call, and returns **authentic / synthetic / uncertain**, a confidence and a **LOW / MEDIUM / HIGH** risk level. It helps an analyst decide whether a high-risk request needs extra verification. It is not a biometric authenticator and not proof of fraud.

Research questions and hypotheses

- **RQ1–RQ4 (Protocol §1):** FoR → DEEP-VOICE generalisation; CNN vs baseline; robustness under degradation; calibration under shift. H1–H7 unchanged.
- **H8 (D48):** model v2 generalises better than v1 to DEEP-VOICE (paired, speaker-clustered bootstrap; Stage 13).
- **H9 (D50):** augmentation transfers to a degradation family never used in training (white noise, 10 dB; Stage 14).

Models

Model	What	Status
`baseline-v1`	MFCC mean + std, RBF-SVM; evaluation only (D7, D37)	Fitted (Stage 7)
v1 CNN	4 conv blocks on F3 log-Mel; seeds 0–2; risk policy on val-B (D42, D45)	Selected and fitted; frozen as F4 at Stage 12
v2 (one of three)	`cnn-aug`, `crnn` or `aasist`, trained with D50 augmentation, selected on val-A (D49)	To build; frozen as F5 by 4 Dec

Deployment rule (D48 (5)): v1 replaces the mock as soon as F4 is verified; v2 replaces v1 once F5 is verified, whatever its test results. Test results never choose the deployed model.

Scope

In scope

- Baseline, v1 CNN, and v2 from the three named candidates
- Risk layer: Platt calibration, EER threshold, uncertain band (≤ 10% deferral on val-B)
- Live analysis: microphone mode, `/api/live`, Chrome tab-capture extension
- FastAPI, React, PostgreSQL, Docker; staging and production
- Robustness (Stage 14) and calibration analysis (Stage 15)

Out of scope (future work)

- XLS-R / wav2vec2 / WavLM / HuBERT / RawNet, ensembles
- Other features (LFCC, CQCC), explainability
- Mobile app; capture of native phone or in-app mobile calls (OS restriction, D51)
- Extra speech datasets; a second external dataset
- Adversarial defence, replay attacks

Metrics (D4)

Positive class = spoof. Score level: EER, ROC-AUC. Decision level at the frozen threshold: accuracy, precision, recall, F1, forced-binary and on non-deferred cases. Always with the 3-way confusion matrix, coverage / deferral, class counts and 95% confidence intervals. Never accuracy alone. No t-DCF.

4. Hard rules

These rules protect the validity of the results. They are in `AGENTS.md`, so every AI tool follows them too, and the most important ones are enforced by tests in CI.

1. **DEEP-VOICE is touched exactly once:** one joint run of v1, v2 and the baseline after F5, or after the 4 December cut-off with v1 and the baseline only (D48, D35). Before it, only the D20 metadata accesses.
2. **Each ModelBundle is frozen as one unit** (checkpoint + preprocessing profile + risk policy), fitted on FoR only, before the external run: F4 for v1, F5 for v2.
3. **Scope lock:** baseline, v1 CNN, and v2 only from `cnn-aug`, `crnn`, `aasist` (D49). Live analysis only as D51–D52.
4. **The FoR test split is read once per model set** (D31, D41), through the reviewed runner, after the D46 gates. The v2 design was fixed before any test result (D48 (6)); a later change is post-hoc and labelled exploratory.
5. **Metrics as in D4.** Never accuracy alone.
6. **No silent changes** to data, splits, preprocessing, thresholds, metrics or scope. Propose, explain the impact, decide.
7. **Never commit audio, datasets, checkpoints or secrets.** Live audio is processed in memory and never stored or logged (D51).

5. How we work

5.1 Git workflow

- `main` is protected: pull request, 1 approval, required checks `checks`, `web` and `e2e`, branch up to date.
- One meaningful unit = one branch = one PR. Branch names `<type>/stage<N>-<name>` (D23); Conventional Commits.
- **Decision PRs:** all four approve in the PR, then the **approval commit** (statuses, AGENTS.md, protocol) goes on the branch **before** merge. Lesson from PR #67, which needed the record PR #112.
- **Numbers** (D-IDs, ADRs, log entries) are taken at merge time; the PR that merges second renumbers.
- After every merge: sync `main`, delete the branch locally and on GitHub, `git fetch --prune`, remove scratch files.

5.2 AI tools and human in the loop

- Claude (claude.ai) writes Claude Code prompts and reviews diffs line by line; each member drives their own tool (Claude Code, Codex, Antigravity). `AGENTS.md` is the single rulebook (D22).
- Ground truth over self-report: a claim counts only with real output (tests, data, hashes). Labels [verified], [doc], [hypothesis].
- AI never merges. AI assistance is recorded in each PR and in the project log.

5.3 Weekly rhythm

When	What
Monday	Weekly sync (30 min): last week's gates, this week's cards, blockers, critical-path check.
Every day	Small PRs; reviews within 24 hours; a blocked card is said the same day.
Friday	Each owner updates the project log for their track; the plan's status column is updated by PR when a gate passes.

5.4 Reproducibility and data

- Every result = pinned data + config file + code commit. Python 3.13 official; exact pins; CI on 3.13 with torch.
- Audio lives on the shared Drive only; datasets and the new noise/RIR corpora are pinned by hash (D21, D50 (5)).
- Every training run is logged to W&B with its config (D40). No audio, test or external data goes to W&B.

5.5 What a gate means

A step is closed when (1) its outputs are merged through a reviewed PR with green CI, (2) its review or sign-off is recorded, (3) the project log has an entry and an updated Current status, and (4) for a stage, its brief is merged and the teach-back is done (D32).

6. Team, ownership and reviews

Member	Owns	This plan's main work	Reviews
Donia Mabrouk	Part C (Stages 7–12) · Track V lead	Stage 11 PR B, PR C, dry run, official run; F4; v2 candidates, training, selection, policy, v2 Stage 11; F5	Malak's V1 pipeline; Stage 14
Malak Ben Salem	Part B (4–6) · V1 · Stage 14	D50 corpora, splits and augmentation module; selection set S; Stage 14 for three model sets	Part E (Iheb)
Khalil Dridi	Stages 13, 15, 16 · SRS v4.0	Stage 13 runner for three model sets (D35, D36, D48); D52 warm-up analysis; Stages 15–16; SRS v4.0 draft	Part C (Donia), required reviewer
Iheb Zemzemi	Part E (17–19) · Tracks L and P · SDD v2.0	Real model in the API; PostgreSQL, process pool, load test; staging/production; live endpoint, live UI, extension; UI redesign	Track V code (with Khalil)
Everyone	Decision PRs · Stage 13 review · Stage 19	Usability test sessions; report sections for own stages; final presentation	—

> **Separation that protects the result.** The person who builds the models (Donia) never runs the external test. Khalil runs Stage 13 once, for all frozen models together; the result cannot be used to change any model, threshold or deployment choice.

Hand-overs

Hand-over	What is fixed in advance
Malak → Donia (V1 → V2)	Augmentation function signature, families, probabilities and seeds (D50); selection set S with hashes.
Donia → Iheb (F4, F5)	ModelBundle format and loader (verified hashes); analyzer interface unchanged; bundle version in every record.
Donia → Khalil (F4, F5)	Frozen bundles, `baseline-v1` hash, per-clip Stage 11 files; stored-score format per model (D36).
Iheb → all (P3)	Staging URL, test accounts, demo audio set; never real personal recordings (D54 (8)).

7. Timeline and critical path

Six weeks plus a short buffer. Weeks run Monday to Sunday. Dates are targets the owner commits to in the Monday sync; the **4 December cut-off is a rule**, not a target (D48 (4)).

Week	Dates	Track R (v1)	Track V (v2)	Part D	Tracks L and P
W1	9–15 Nov	PR B, PR C	V1 corpora pinned, augmentation module	Stage 13 runner: three model sets	UI directions and vote; PostgreSQL; process pool; hosting chosen
W2	16–22 Nov	Dry run → official Stage 11 → F4	V2 candidate code; V3 training starts	Runner review	Real model in the API (after F4); staging live; UI build starts
W3	23–29 Nov	Stage 11 results, brief, teach-back	V3 selection; V4 policy fit	D52 warm-up analysis (val-B)	`/api/live` and live mic mode; extension prototype
W4	30 Nov–6 Dec	—	V4 v2 Stage 11; **V5 F5 by Fri 4 Dec**	**Stage 13 joint run (Sat 5–Sun 6 Dec)**	Extension on WhatsApp Web; load test → limits
W5	7–13 Dec	—	—	Stages 14, 15, 16	Production deploy; usability test → fixes
W6	14–20 Dec	Report	Report	Report	Report, demo set, two rehearsals
Buffer	21–23 Dec				Final checks; submission by Wed 23 Dec (target)

Critical path to the cut-off

Step	Owner	Target date	Depends on
Noise and RIR corpora: licence, hash, 50/50 split pinned	Malak	Fri 13 Nov	—
Augmentation module + selection set S merged	Malak	Wed 18 Nov	corpora
v2 candidate code (`cnn-aug`, `crnn`, `aasist`) merged; batch-size benchmark in `configs/`	Donia	Fri 20 Nov	augmentation module
Training (seed 0 per candidate) and selection on val-A	Donia	Fri 27 Nov	candidate code
Seeds 1–2 of the winner; v2 risk policy on val-B	Donia	Mon 30 Nov	selection
Runner accepts the v2 model set; v2 Stage 11 run (once)	Donia	Wed 2 Dec	policy
F5 verified by the bundle loader	Donia	**Fri 4 Dec, 23:59**	v2 Stage 11
Stage 13 joint run (v1, v2, baseline)	Khalil	Sat 5–Sun 6 Dec	F5 or cut-off; Stage 13 runner

> **Cut line, if a week slips.** Cut in this order: (1) the Chrome extension: live mic mode still covers live analysis; (2) the `aasist` candidate: `crnn` still answers the temporal-model request; (3) the usability test shrinks to 3 participants. Never cut: the real model in the platform, the single-run rules, augmentation, Stage 11 and Stage 13.

8. Track R — v1 to its freeze (Stages 11–12) · Donia

11 FoR internal test, v1 (once) — NOW

DONE D41 registry and per-clip storage (PR #62); D46 procedure (PRs #67, #112); runner, PR A (PR #68).

TASKS

- **PR B:** report computed only from the stored per-clip files: D4 metrics with class counts per subgroup, H1 with bootstrap CI, H5, seed mean ± SD.
- **PR C:** notebook 06: environment gate and record, reproduction gate, rehearsal, claim, Drive flush and remount, registry SHA-256 against `claim.json`, `run_id` from `claim.json` in code, score, export.
- **Dry run on T4**, validation splits only (MFCC-on-T4 still unverified), then the freeze checklist.
- **Official run once**; results PR; registry committed byte for byte (D46 (6)); brief and teach-back.

DATES PR B Fri 13 Nov · PR C Mon 16 Nov · dry run Tue 17 Nov · official run Wed 18–Thu 19 Nov.

GATE One completed `stage11_test` claim for the v1 model set; results and brief merged; Khalil's teach-back.

12 ModelBundle v1 freeze (F4) and hand-over — NEXT

TASKS

- Assemble F4: bundle-seed checkpoint + F3 profile + `risk-v1` + manifest; record `baseline-v1` hash (D36).
- Verify with the bundle loader; hand over to Iheb for the API (P1) and to Khalil for Stage 13.

DATES F4 verified Sun 22 Nov.

GATE Loader verifies every part; F4 manifest SHA-256 recorded in the log; the mock can be removed.

9. Track V — model v2 (V1–V5) · Donia, with Malak

V1 Corpora and augmentation pipeline (D50) — NEXT

OWNER Malak; review Donia.

TASKS

- Noise (MUSAN, OpenSLR 17) and RIR (RIRS_NOISES, OpenSLR 28): licence, download hash, seeded 50/50 split by file into training and selection halves; pins in `configs/`.
- Augmentation module on the waveform before F3: five families with the D50 values; REAL and FAKE alike; probability 0.8; no white noise; no §9 MP3 bitrates or low-pass family; ffmpeg and codecs pinned.
- Selection set S built once from val-A (seed 2026): clean, noise 10 dB, reverberation, Opus 16 kbps, narrowband; hashes stored.
- Tests on synthetic audio: determinism with a seed, family probabilities, peak limit, disjointness checks.

DATES Corpora Fri 13 Nov · module and S merged Wed 18 Nov.

GATE Merged with tests; pins and hashes recorded; Donia's review.

V2 Candidate models — TO DO

TASKS

- `cnn-aug` reuses the v1 network (D42); `crnn` and `aasist` as specified in D49 (`aasist` from a reference implementation pinned by commit, F3 waveform input).
- Unit tests (shapes, one logit, determinism); short compute benchmark → batch sizes in `configs/` before training.

DATES Merged Fri 20 Nov. Review Khalil and Iheb.

V3 Training and selection on val-A — TO DO

TASKS
- Seed 0 per candidate, max 30 epochs, early stopping on the mean val-A EER over S (patience 5); W&B for every run.
- Winner = lowest mean; ties: fewer parameters, then `cnn-aug`, `crnn`, `aasist`. Seeds 1 and 2 for the winner; bundle seed = median.

DATES Selection Fri 27 Nov · seeds 1–2 Mon 30 Nov.

GATE Selection results committed (validation only); every candidate's val-A table reported.

V4 v2 risk policy and v2 Stage 11 — TO DO

TASKS
- Protocol §6 method unchanged on clean val-B, bundle seed only (D45 rule).
- Runner extended to the v2 model set (small PR); one `stage11_test` claim for v2, all three seeds stored.

DATES v2 Stage 11 Wed 2 Dec.

V5 ModelBundle v2 freeze (F5) — TO DO

TASKS
- Assemble and verify F5 with the existing loader; hand over to Iheb (deployment) and Khalil (Stage 13).

DATES **Fri 4 Dec, 23:59 Tunis: cut-off (D48 (4)).**

GATE F5 verified before the cut-off, or v2 is excluded from Stage 13 and that is logged.

10. Part D — joint external run and analyses (Stages 13–16) · Khalil, Malak

13 DEEP-VOICE external run (once, guarded) — TO DO

PREPARE
- Stage 13 runner (D35) for three model sets in one pass: v1 bundle CNN, v2 bundle model, `baseline-v1`; one stored-score file per model with identical columns (D36); D17 filter, D19 per-speaker table, D33, D34 file-level view.
- Governance allowlist decision before the run (Protocol §8 open item); durable claim beyond Colab (D35).
- D52 warm-up analysis on simulated calls from val-B (descriptive).

RUN Sat 5–Sun 6 Dec, after F5 or the cut-off. Review by all four.

REPORT H2, H8 (paired v1–v2), RQ2 (CNN vs baseline), the live rule's end-of-call accuracy via D34, 87.5% FAKE share and RVC-only scope stated.

14 Robustness (FoR test only) — TO DO

OWNER Malak; review Donia.

TASKS
- Complete §9 grid for `baseline-v1`, v1 bundle CNN and v2 bundle model (one `stage14_robustness` claim each), reusing each model's Stage 11 clean scores (D41).
- For v2, label conditions: white noise = unseen family, MP3 = seen family / unseen bitrates, bandwidth = seen family (D50 (3)). H4, H6 (v1), H9.

DATES v1 from Mon 23 Nov (after Stage 11); v2 from Mon 7 Dec.

15–16 Calibration and error analysis — TO DO

OWNER Khalil; review Donia.

TASKS
- Reliability diagrams and ECE (10 bins) on FoR test and stored DEEP-VOICE scores; H3; deferral under shift (RQ4).
- Errors by subgroup, speaker, confidence and deferral; H5, H7, D43 (v1; descriptive for v2).

DATES Mon 7–Sun 13 Dec.

11. Track L — live analysis (L1–L3) · Iheb

L1 Streaming endpoint `/api/live` — TO DO

TASKS

- WebSocket through nginx, session required; mono PCM at native rate; server-side F3 including `soxr_hq` (D52 (1)).
- Non-overlapping 2 s windows; D17 speech filter; per-window calibrated p; running verdict = D34 rule on the stream; warm-up 3 kept windows; trend of the last 5 windows without a label.
- Same process pool and limits as uploads (D53); per-message size cap; no audio stored or logged; one metadata record per session.

DATES Built against the mock W2–W3; real bundle after F4.

L2 Live microphone mode (web app) — TO DO

TASKS

- Start/Stop, visible listening indicator, consent and decision-support notice (D51); live screen of the chosen design (D55).

DATES W3–W4.

L3 Chrome extension (Should) — TO DO

TASKS

- Design first: authentication against D47 (SameSite=Strict cookie, Origin check) decided and security-reviewed in SDD v2.0.
- Manifest V3, `tabCapture` after a user click, offscreen document, tab audio routed back to the speakers; same endpoint; tested on WhatsApp Web and Google Meet.

DATES Prototype W3; WhatsApp Web W4. First item on the cut line.

12. Track P — platform and production (P1–P5, Stage 19) · Iheb

P1 Real model in the API — TO DO

TASKS

- Replace `MockAnalyzer` with the verified F4 bundle behind the existing analyzer interface; versions in every record (NFR-07, NFR-09).
- End-to-end tests with the real bundle; later swap to F5 by the deployment rule (D48 (5)).

DATES Tue 24 Nov (after F4).

P2 PostgreSQL and process pool (D53) — NOW

TASKS

- PostgreSQL behind the repository interface for analyses, notes and D47 sessions; migration; CI service container.
- Bounded process pool (kills timed-out work); `503` + `Retry-After` when full; no broker.

DATES W1–W2.

P3 Staging and production (D54) — TO DO

TASKS

- Hosting provider chosen and recorded (W1); images to GHCR once per `main` commit; staging auto-deploy; production from a release tag with a required reviewer; HTTPS; smoke test and rollback.

DATES Staging live W2; production W5.

P4 **Load test and limits (D53 (3))** TO DO

TASKS
- Reference workload on staging with the deployed bundle; record K, L, `analysis_timeout_s`, `L_MAX` by decision; tmpfs follows K.

DATES W4.

P5 **Interface redesign and usability (D55)** NOW

TASKS
- Three directions (Claude prepares) → team vote → implementation keeping §11 fields, NFR-05 wording, axe scans.
- Decision board; per-window timeline; live screen.
- Moderated test, 3–5 people outside the team, fixed task script, SUS ≥ 68; findings and fixes logged.

DATES Vote by Fri 13 Nov · build W2–W4 · test W5.

19 **Integration, report and delivery** TO DO

TASKS
- SRS v4.0 (Khalil, W2) and SDD v2.0 (Iheb, W3); README and setup current; fresh-machine test (NFR-10).
- Final report: each owner writes their stages; demo audio set; two full rehearsals.

DATES W5–W6; submission target Wed 23 Dec.

13. Risks and mitigations

Risk	Mitigation
v2 training does not fit the free T4 time	One seed per candidate during selection; 30-epoch cap; batch benchmark in W1; cut line drops `aasist` first; cut-off protects Stage 13.
Stage 11 reproduction gate fails (feature bytes differ)	Dry run first on validation splits; a mismatch stops Stage 11 for a logged decision before any test access (D46 (4)).
Test results tempt a change to v2	v2 design approved before any test result (D48 (6)); deployment by rule (D48 (5)); registry refuses reruns.
Iheb's tracks are the heaviest	Cut line starts with the extension; Khalil writes SRS v4.0; UI directions prepared by Claude; reviews spread.
Extension cannot authenticate safely	Design and security review before code (D51); live mic mode remains the Must.
Merge before approval commit	Approval commit on the branch before merge (Section 5.1); record PR otherwise, as #112.
AI-generated errors accepted unchecked	Line-by-line review, real test output, CI with torch; ground truth over self-report.

14. Open items this week (W1)

1. Monday sync: confirm every date in Section 7 and the delivery date against the course calendar.
2. Donia: Stage 11 PR B (Claude Code prompt), then PR C.
3. Malak: D50 corpora licence, hashes and split; augmentation module (Claude Code prompt).
4. Khalil: Stage 13 runner for three model sets; start the SRS v4.0 outline.
5. Iheb: hosting provider recorded; PostgreSQL and process pool; review of the three UI directions.
6. All: vote on a UI direction by Friday 13 November.
7. Adopt this plan: one docs PR with this file and its PDF in `docs/reference/`.

Appendix A — Decisions since v2.1

Authoritative text: `docs/decisions/README.md`. Decisions D1–D25 are summarised in v2.1, Appendix A.

ID	Decision
D26	Work distribution (Execution Plan v2.1): parts A–E, owners, consumer reviews producer.
D27	Shared AI tooling: one skill (`pr-prep`) and one MCP server (Context7).
D28	Dataset freeze F2: FoR manifest and hash ledger.
D29–D31	Baseline threshold at its own val-B EER point; CNN search space; Stage 11 runs once (registry).
D32	Stage briefs and teach-back are part of stage closure.
D33–D36	Stage 13 reporting conventions, file-level view, guarded access, baseline in the single external run.
D37–D40	Baseline tuning; React replaces Streamlit (ADR 0008); F3 preprocessing profile; W&B from Stage 8.
D41	Registry purpose key; paired clean scores for Stage 14 (ADR 0010).
D42–D45	CNN design; loudness diagnostic; Platt stop rule; calibrated output for the bundle seed only.
D46	Stage 11 run procedure: T4, CPU scoring, environment gate and record, reproduction gate, durable claim, rerun rule.
D47	Analyst sign-in: username + password, server-side cookie sessions (ADR 0011).
D48	v2 track; one joint external run; cut-off 4 Dec; deployment by rule (ADR 0012).
D49–D50	v2 candidates and selection; training augmentation disjoint from Stage 14 (ADR 0013).
D51–D52	Live analysis scope; live decision rule = D34 on the stream (ADR 0014).
D53–D54	CPU process pool, PostgreSQL, measured limits; dev/staging/production (ADR 0015).
D55	Interface redesign and usability evidence.
H7–H9	Pauses (Stage 16); v2 vs v1 on DEEP-VOICE (Stage 13); unseen white-noise family (Stage 14).

Appendix B — Freeze points and registry claims

Freeze / claim	Content	When
F1	Experimental Protocol v1.0 (+ addendum, PR #111)	Stage 3 (done)
F2	Split, manifests and hash ledger (D28)	Stage 5 (done)
F3	Preprocessing profile (D39)	Stage 6 (done)
F4	ModelBundle v1 + `baseline-v1` hash (D36)	Stage 12, target Sun 22 Nov
F5	ModelBundle v2 (D49)	V5, by Fri 4 Dec 23:59
`stage11_test`	One claim per model set: v1 set, then v2 set (D31, D41, D48 (7))	W2 and W4
`stage14_robustness`	One claim per model set: baseline, v1, v2 (D41, D48 (8))	W3–W5
Stage 13	One dataset-wide claim: all frozen models in one pass (D35, D36, D48)	Sat 5–Sun 6 Dec

Appendix C — Document versions

Document	Version now	Next	Owner, target
Decision register	D1–D55	New rows as needed	Decision owner

Document	Version now	Next	Owner, target
Experimental Protocol	v1.0 + addendum (PR #111)	Further addenda only by decision	Donia
SRS	v3.0	v4.0: live analysis (WH-01 changed), infrastructure, priorities	Khalil, W2
SDD	v1.0	v2.0: streaming, process pool, PostgreSQL, environments, extension security	Iheb, W3
Execution Plan	v3.0 (this)	Status updates by PR	Donia

Appendix D — What changed since v2.1

v2.1 said	v3.0 does
One deployed CNN; augmentation out of scope (D10)	v1 CNN kept; v2 from three candidates with augmentation (D49, D50).
Stage 13 right after the v1 freeze	Stage 13 waits for F5 or the 4 Dec cut-off; one joint run (D48).
Streamlit dashboard	React + TypeScript web app (D38); redesign and usability test (D55).
Upload-only service; live detection out of scope	Live mic mode and `/api/live`; Chrome extension (D51, D52).
SQLite; one environment	PostgreSQL, process pool, measured limits; staging and production (D53, D54).
14-week targets	Dated six-week plan with a critical path and cut line.
Khalil (surname not recorded)	Khalil Dridi; owns SRS v4.0 and the joint Stage 13 run.

Prepared from the decision register (D1–D55), Protocol v1.0 and its addendum, the project log (Entries 1–154), Execution Plan v2.1, the v2.2 draft, SRS v3.0 and SDD v1.0, with AI assistance (Claude); to be reviewed by the team.